DESCRIPTIVE STATISTICS · STUDY NOTES

Correlation and the Correlation Coefficient

How two variables move together, and how one number between −1 and 1 measures it

30-SECOND REVISION

- **Formula:** r = covariance ÷ (SD of x × SD of y)
- **Range:** always between −1 and +1, with no units
- **Meaning:** +1 perfect positive · 0 no linear relationship · −1 perfect negative
- **Class results:** housing r = 0.87; SAT r = 0.94
- **Warning:** correlation does not imply causation

Learning objectives

After these notes you should be able to:

- define **correlation** and the **correlation coefficient**, and explain how they are related;
- calculate r from the covariance and the two standard deviations;
- interpret the sign and the size of r;
- explain why correlation does not imply causation.

Contents: 1 What is correlation? · 2 From covariance to the correlation coefficient · 3 The correlation coefficient · 4 Worked example: housing · 5 Properties of r · 6 Correlation and causation · 7 Exercise: SAT scores · 8 Excel and Python · 9 Revision kit

Items marked *(added)* are not from the class notes; they were added to make the topic clearer.

1. What is correlation?

Definition: correlation

Correlation is a statistical relationship between two variables. It describes whether the variables tend to change together and, if they do, in which direction. If two variables are correlated, a change in one is usually accompanied by a change in the other.

Example. In real estate, larger houses tend to cost more, so house size and house price are correlated. A scatter plot with size on the x-axis and price on the y-axis shows this relationship clearly.

Types of correlation

Type	What it means	Example
Positive	Both variables tend to rise together	House size and house price
Negative	One variable tends to rise while the other falls	Ice-cream sales and umbrella sales
None	No pattern links the two variables	Coffee production in Brazil and house prices in London



Figure 1. Illustrative scatter plots (not class data) of the three types of correlation.

Correlation and the correlation coefficient

The two terms are closely related but they are not the same thing. **Correlation** is the relationship itself. The **correlation coefficient (r)** is the number that measures that relationship. As an analogy, temperature is a property of the air and a thermometer reading in °C is its measurement.

	Correlation	Correlation coefficient (r)
What it is	A concept: how two variables move together	A statistic: one number that measures that relationship
Questions it answers	Do the variables move together? In which direction?	In which direction, and how strongly?
How it is shown	In words, or as a pattern on a scatter plot	As a number from −1 to +1
Example	“House size and price are positively correlated.”	“r = 0.87”

Note on terminology
In everyday speech people often say “the correlation is 0.87” when they mean “the correlation coefficient is 0.87”. In these notes, **correlation coefficient** always means the **linear (Pearson) correlation coefficient**. This is the standard measure, and it is what Excel's `CORREL` and `PEARSON` functions calculate. It measures straight-line relationships only.

2. From covariance to the correlation coefficient

The previous topic ended with one problem: a covariance such as **33,491,250** tells you the **direction** of a relationship (positive), but you cannot tell whether it is strong or weak. The **correlation coefficient** adjusts the covariance so the relationship becomes easy to interpret.

Covariance
direction only; its size depends on the units

$\div (s_x \cdot s_y)$ →

Correlation coefficient
direction and strength; no units

	Covariance	Correlation coefficient
Tells you the direction	Yes (by its sign)	Yes (by its sign)
Tells you the strength	No	**Yes**
Units	unit of x × unit of y	**None**
Range	any number	**−1 to 1**
Easy to compare across data sets	No	**Yes**

3. The correlation coefficient

> **Definition: correlation coefficient (r)**
> A number, calculated from data, that measures the **direction and strength of the linear relationship** between two variables. It equals the covariance divided by the product of the two standard deviations.

The correlation coefficient has two symbols, depending on the data it is calculated from. Both mean the same thing: the direction and strength of the linear relationship.

Symbol	Name	Calculated from
r	Sample correlation coefficient	A sample (it estimates ρ)
ρ (rho)	Population correlation coefficient	The whole population

Use sample values throughout or population values throughout. The number comes out the same either way, because the n − 1 (or N) cancels. The examples in these notes are samples, so their results (0.87 and 0.94) are values of r.

$$r = \frac{\text{cov}(x, y)}{s_x s_y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})/(n-1)}{s_x s_y}$$

$$\rho = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

How to read the value

r	Name	Meaning
+1	Perfect positive correlation	All points lie exactly on a rising straight line
Between 0 and +1	Imperfect positive	x and y tend to rise together; closer to 1 is stronger
0	No linear correlation	No straight-line relationship between x and y
Between −1 and 0	Imperfect negative	One tends to rise as the other falls; closer to −1 is stronger
−1	Perfect negative correlation	All points lie exactly on a falling straight line

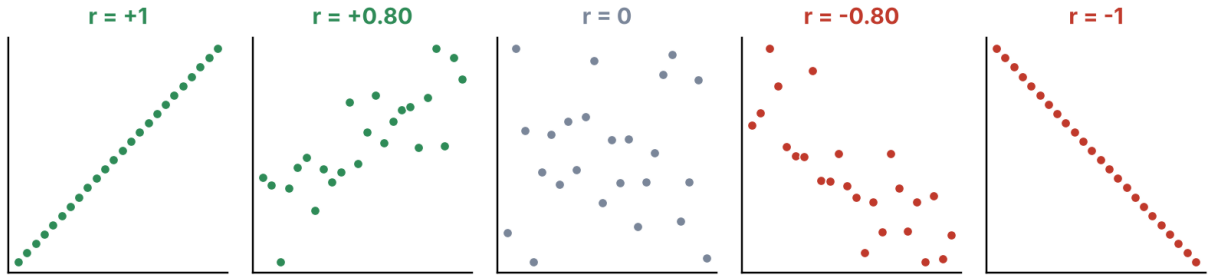


Figure 2. Illustrative scatter plots (not class data) showing how r changes with the pattern of the points.

Key fact
The correlation coefficient can **never** be greater than 1 or less than −1. If a calculation gives 1.3 or −2, it contains an error.

4. Worked example: housing data

These are the same five houses as in the covariance topic. Because they are only part of all houses, they form a **sample**.

4.1 The data set

Housing: Size vs Price

DATA SET
EXAMPLE

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Data set example · 25.0 KB · 3 sheets

Housing data: 5 houses with their size and price

Attached inside this PDF. Click the file icon, or open your PDF viewer's Attachments panel. Keep a copy beside this PDF in the repo folder as well.

House	Size (ft.)	Price ($)
1	650	$ 772,000
2	785	$ 998,000
3	1,200	$ 1,200,000
4	720	$ 800,000
5	975	$ 895,000

4.2 The steps

Step 1: covariance. From the previous topic: cov(size, price) = 133,965,000 ÷ (5 − 1) = **33,491,250**.

Step 2: standard deviation of size. Sum of squared distances = 197,970. Divide by n − 1 = 4 to get the variance (49,492.50), then take the square root: s_x = **222.47** ft.

Step 3: standard deviation of price. Sum of squared distances = 120,568,000,000. Divide by n − 1 = 4 to get the variance (30,142,000,000), then take the square root: s_y = **$173,614.52**.

Step 4: divide. r = 33,491,250 ÷ (222.47 × 173,614.52) = **0.8671** ≈ **0.87**.

The building blocks

House	x − 866	$(x - \bar{x})^2$	y − 933,000	$(y - \bar{y})^2$ (billions)	$(x - \bar{x})(y - \bar{y})$
1	−216	46,656	−161,000	25.921	+34,776,000
2	−81	6,561	+65,000	4.225	−5,265,000
3	+334	111,556	+267,000	71.289	+89,178,000
4	−146	21,316	−133,000	17.689	+19,418,000
5	+109	11,881	−38,000	1.444	−4,142,000
Sum		**197,970**		**120.568**	**133,965,000**

$$r = \frac{33{,}491{,}250}{222.47 \times 173{,}614.52} = 0.87$$

What the plot shows

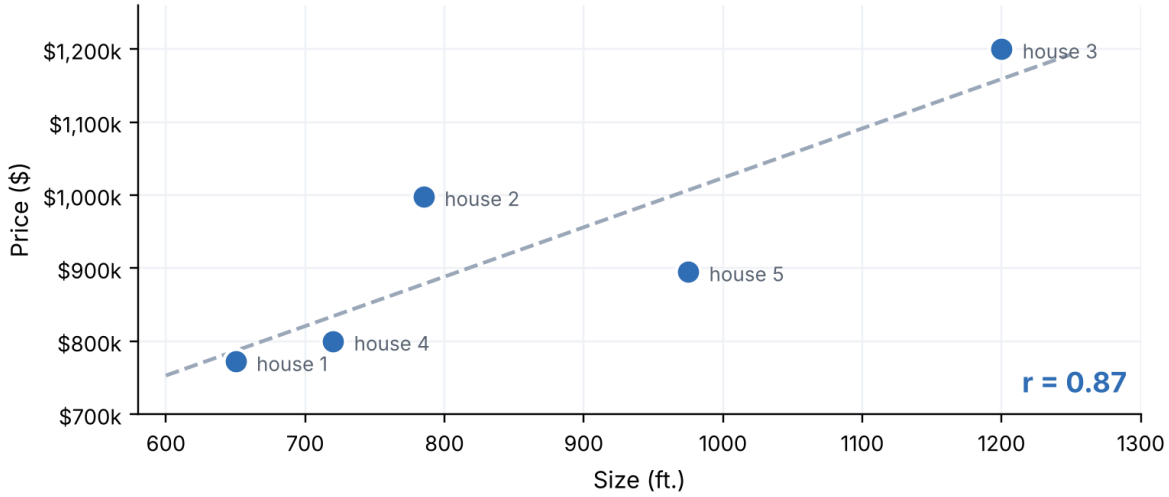


Figure 3. Housing data. The points follow a rising line closely, which is why r is high. The dashed line is the best-fit straight line (added for illustration).

Result
r = **0.87**. The correlation is **strong and positive**: larger houses tend to cost more.
Note: r is a pure number with no units. It does **not** mean that each extra foot adds $870 to the price. The price gained per foot is the slope of the fitted line: cov ÷ variance of size = 33,491,250 ÷ 49,492.50 ≈ **$677** per foot (added).

How much does size explain? (added)
Squaring r gives $r^2 = 0.87^2 \approx$ **0.75**: about **75%** of the variation in price is linked to size through a straight-line relationship. The remaining 25% comes from other factors (location, age, condition). So size is the main driver here, but not the only one.

5. Properties of the correlation coefficient

- **Between −1 and +1.** It is always in this range, which makes it easy to interpret.
- **Unit-free.** The units of x and y cancel when you divide the covariance by the two standard deviations.
- **Symmetric.** The correlation between x and y equals the correlation between y and x. The formula is completely symmetric: swapping the variables gives 0.87 again.
- **Not changed by rescaling.** Change dollars to thousands of dollars, or feet to metres, and r stays the same. The covariance does not. (The charts below show this for the housing data.)

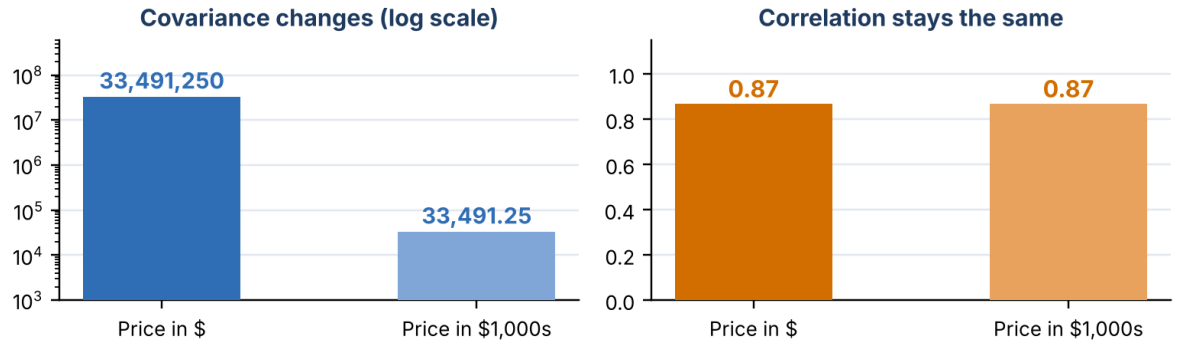


Figure 4. Changing the price from dollars to thousands of dollars changes the covariance but not r (calculated from the housing data).

Perfect correlation, in numbers (added)

x	y	Pattern	r
1, 2, 3	2, 4, 6	y rises by 2 for every 1 of x	**+1**
1, 2, 3	6, 4, 2	y falls by 2 for every 1 of x	**−1**

r = 1 in the real-estate example
The class example: if every extra square foot added the same amount to the price (say 1 sq. ft. = $1,000, so 5 sq. ft. = $5,000), all houses would lie on one straight line and r would be exactly **1**. Real data never behaves that perfectly, which is why r = 0.87 is not 1.

6. Correlation and causation

Correlation shows **whether** two variables move together. It says nothing about **why**.

Idea	What the class notes say
Direction of cause	In the house business, **size causes price**, not the other way round: you decide the size when you build, and a bigger house costs more. Raising the price of an existing house does not make it bigger.
Why it matters	Causality means knowing which variable is the cause and which is the effect. Correlation is symmetric, so it cannot tell you this.
Conclusion	**Correlation does not imply causation.**

Class examples of zero and negative correlation

Case	Example	What happens
r = 0	Coffee production in Brazil and house prices in London	The two variables have nothing in common, so no relationship shows up.
r < 0	Ice-cream sales and umbrella sales	Ice cream sells more in good weather and umbrellas sell more when it rains, so when one rises the other falls.

Two warnings (added)
1. r = 0 means no *linear* relationship, not "no relationship". The points x = −2, −1, 0, 1, 2 and y = 4, 1, 0, 1, 4 follow $y = x^2$ perfectly, yet r = 0 (see Figure 5).
2. A strong r can come from a third factor. Ice cream and umbrellas are both driven by the weather, not by each other. Always ask what could explain the link before claiming a cause.

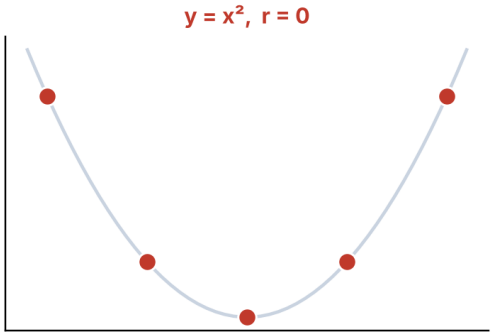


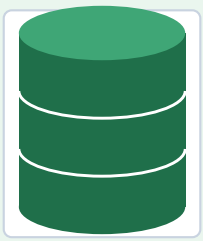
Figure 5. A perfect curved relationship ($y = x^2$) can still have r = 0.

7. Exercise: SAT scores

Data on SAT writing and reading scores from the class exercise. These are the same five students as in the covariance exercise.

Task	What to do	Answer
1	Calculate the correlation coefficient of the two data sets	**r = 0.94**
2	Comment on the strength of the correlation	**Very strong positive** (0.94 is extremely close to 1)

7.1 The data set


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SAT Scores: Writing vs Reading

DATA SET
EXAMPLE

Data set example · 29.5 KB · 3 sheets
Exercise and solution: 5 students, SAT writing and reading

Attached inside this PDF. Click the file icon, or open your PDF viewer's Attachments panel. Keep a copy beside this PDF in the repo folder as well.

Student	Writing (x)	Reading (y)
1	344	378
2	383	349
3	611	503
4	713	719
5	536	503

7.2 Solution, step by step

Quantity	Writing (x)	Reading (y)
Mean	517.4	490.4
Variance (÷ n − 1 = 4)	23,874.3	21,300.8
Standard deviation	154.51	145.95
Covariance (÷ n − 1 = 4)	21,155.55	

$$r = \frac{21{,}155.55}{154.51 \times 145.95} = 0.94$$

Task 1. Divide the covariance by the two standard deviations: r = **0.94**. In the class workbook, Excel's built-in CORREL function (cell G20) gives the same value; the covariance is calculated by hand in cell G19.

Task 2. r = 0.94 is extremely close to 1, so there is a **very strong positive correlation**: students with higher writing scores almost always have higher reading scores.

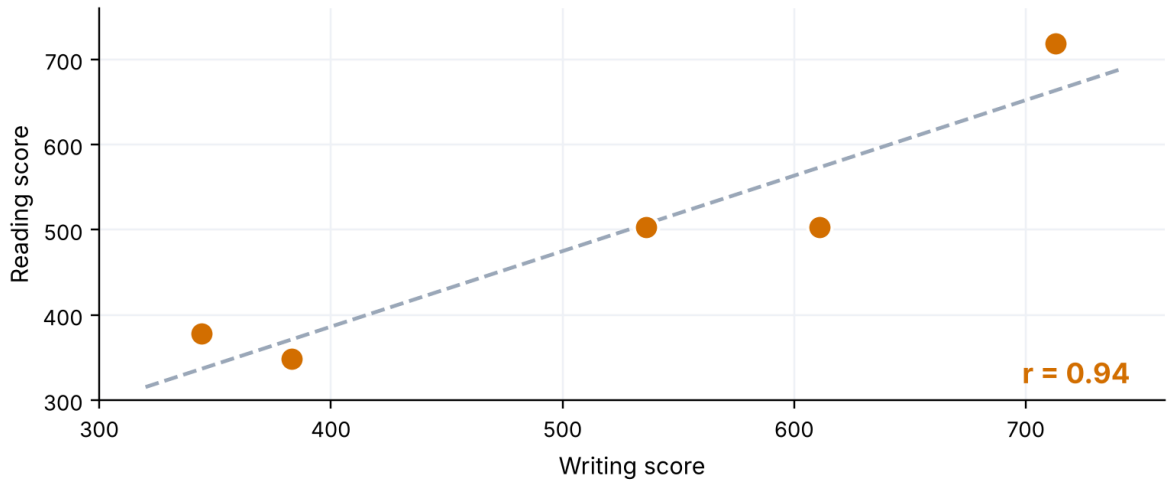


Figure 6. SAT scores. The dashed line is the best-fit straight line (added for illustration).

Compare the two examples (added)
The SAT correlation (**0.94**) is higher than the housing one (**0.87**), so the SAT relationship is the stronger one, even though its covariance (21,155.55) is far smaller than the housing covariance (33,491,250). Covariance alone could not make this comparison. Squaring: SAT $r^2 \approx 0.88$, housing $r^2 \approx 0.75$.

8. Excel and Python

Goal	Excel	Result (housing)
Correlation coefficient (built-in)	=CORREL(C6:C10, D6:D10)	0.8671
Same, alternative function	=PEARSON(C6:C10, D6:D10)	0.8671
Built by hand, as in the workbook	=G13/C12/D12	0.8671
Standard deviation (sample)	=STDEV.S(C6:C10)	222.47

```
import statistics as st
import numpy as np
import pandas as pd

size  = [650, 785, 1200, 720, 975]
price = [772000, 998000, 1200000, 800000, 895000]

st.correlation(size, price)             # 0.8671128624942824  (Python 3.10+)
np.corrcoef(size, price)[0, 1]          # 0.8671128624942823  (2 x 2 matrix: take [0, 1])
pd.Series(size).corr(pd.Series(price))  # 0.8671128624942824
```

Good to know
Because the n − 1 (or N) cancels out, the correlation is the same whether you use sample or population covariance and standard deviations, as long as you use them consistently. `np.corrcoef` returns a 2 × 2 matrix, so pick the off-diagonal entry `[0, 1]`.

9. Revision kit

Work through this section in order: remember the key points, check the formulas, read the scale, test yourself and avoid the common mistakes.

9.1 Five things to remember

1	**Formula.** r = cov(x, y) ÷ (s_x × s_y).
2	**Range.** r is always between −1 and +1, and has no units.
3	**Sign and size.** The sign gives the direction. The distance from 0 gives the strength.
4	**Class results.** Housing r = 0.87 (strong positive). SAT r = 0.94 (very strong positive).
5	**Two warnings.** r = 0 means no *linear* relationship. Correlation does not imply causation.

9.2 Formulas

Quantity	Formula
Correlation coefficient	r = cov(x, y) ÷ (s_x · s_y)
Sample covariance	cov(x, y) = Σ(x − x̄)(y − ȳ) ÷ (n − 1)
Sample standard deviation	s = √[Σ(x − x̄)² ÷ (n − 1)]
Share explained (added)	r² (for example 0.87² ≈ 0.75, so about 75%)

9.3 Reading r at a glance

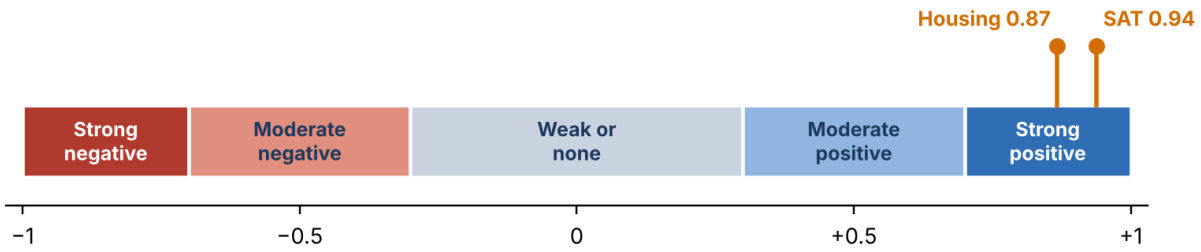


Figure 7. The labels are a common rule of thumb (added), not a strict rule. Judge r together with the scatter plot.

9.4 Self-test (cover the answers and try first)

Concept check

#	Question	Answer
1	Why use r instead of the covariance?	r has no units and lies between −1 and 1, so its size is easy to interpret.
2	Between which values can r lie?	From −1 to 1, including both ends.
3	What does r = 0.87 tell you about house size and price?	A strong positive linear relationship. It does **not** mean \$870 more per foot.
4	If r = 0, are x and y independent?	Not necessarily. r = 0 only rules out a *linear* relationship (for example $y = x^2$).

Practice

#	Question	Answer
5	Find r for x = 1, 2, 3 and y = 6, 4, 2.	The points lie on a falling straight line, so **r = −1**.
6	Ice-cream sales and umbrella sales have r < 0. Does one cause the other?	No. Both depend on the weather. Correlation does not imply causation.

9.5 Common mistakes

Mistake	Correct idea
Reading r as a percentage or as a slope (“\$870 per foot”)	r has no units. The slope is cov ÷ variance of x.
Saying that correlation proves causation	Correlation only shows that the variables move together.
Treating r = 0 as “completely unrelated”	The relationship may be curved. r = 0 only rules out a straight-line link.
Judging strength from the covariance	Covariance depends on units. Use r.
Mixing sample and population values in one formula	Use sample values throughout or population values throughout.

Source: class notes (Part 6), *02_Correlation.xlsx* (attached on page 4) and *03_Correlation_Example.xlsx* (attached on page 7). Items marked *(added)*, such as the definitions in section 1, the scatter gallery, the r^2 explanation and the self-test, were added for clarity. All numbers were checked by computation against the workbooks.